

Organic Chemistry

1. Introduction

Organic chemistry is the branch of chemistry that deals with carbon compounds, especially those containing carbon-hydrogen bonds and their derivatives. It is central to BSc Chemistry because it explains the structure, properties, reactions, and synthesis of compounds found in fuels, medicines, polymers, food, and living systems.

This chapter covers aliphatic hydrocarbons, major functional groups, and biomolecules, which together form the foundation for understanding organic reactivity and biological chemistry.

2. Classification of organic compounds

Organic compounds can be classified based on carbon skeleton and functional groups. Hydrocarbons contain only carbon and hydrogen, while functionalized compounds contain additional atoms or groups that determine chemical behavior.

Main classes

- Aliphatic hydrocarbons.
- Aromatic hydrocarbons.
- Functional group compounds.
- Biomolecules.

3. Aliphatic hydrocarbons

Aliphatic hydrocarbons are open-chain or non-aromatic hydrocarbons. They may be saturated or unsaturated.

Types

- Alkanes: saturated hydrocarbons with single bonds.
- Alkenes: contain one or more double bonds.
- Alkynes: contain one or more triple bonds.

4. Alkanes

Alkanes are saturated hydrocarbons with general formula C_nH_{2n+2} for open-chain compounds. All carbon atoms are sp^3 -hybridized and bond angles are approximately tetrahedral.

Nomenclature

- Methane.
- Ethane.
- Propane.
- Butane.
- Pentane.

Properties

- Nonpolar.
- Low boiling points compared to polar compounds.
- Insoluble in water.
- Combustible.

5. Alkenes

Alkenes are unsaturated hydrocarbons containing at least one carbon-carbon double bond. Their general formula is C_nH_{2n} for open-chain monoalkenes.

Structure

The double bond has one sigma bond and one pi bond. The carbon atoms are sp^2 -hybridized and the molecule is planar around the double bond.

Preparation

- Dehydration of alcohols.
- Dehydrohalogenation of alkyl halides.
- Partial hydrogenation of alkynes.

6. Alkynes

Alkynes are unsaturated hydrocarbons containing at least one carbon-carbon triple bond. Their general formula is C_nH_{2n-2} for open-chain monoalkynes.

Structure

The triple bond contains one sigma bond and two pi bonds. Carbon atoms are sp -hybridized and the geometry is linear.

Preparation

- Double dehydrohalogenation.
- Alkylation of acetylides.
- Industrial methods.

7. Conformational analysis of hydrocarbons

Conformational analysis deals with different spatial arrangements of atoms produced by rotation around single bonds. In ethane and butane, staggered and eclipsed forms explain stability differences.

Important idea

More stable conformations have lower torsional strain and steric repulsion.

8. Functional groups

A functional group is an atom or group of atoms that determines the characteristic reactions of an organic compound. Functional groups are the most important feature in organic chemistry because they control polarity, acidity, basicity, and reactivity.

Major functional groups

- Halo group: $-X$.
- Hydroxyl: $-OH$.
- Ether: $-O-$.
- Aldehyde: $-CHO$.
- Ketone: $>C=O$.
- Carboxylic acid: $-COOH$.
- Ester: $-COOR$.
- Amine: $-NH_2$, $-NHR$, $-NR_2$.
- Amide: $-CONH_2$.
- Nitrile: $-CN$.
- Nitro: $-NO_2$.
- Thiol: $-SH$.

9. Alcohols

Alcohols contain the hydroxyl group attached to carbon. They are classified as primary, secondary, or tertiary depending on the carbon bearing the OH group.

Properties

- Hydrogen bonding.
- Higher boiling points than hydrocarbons.
- Soluble in water for lower members.

Reactions

- Oxidation.
- Dehydration.
- Esterification.
- Substitution.

10. Ethers

Ethers contain an oxygen atom between two alkyl or aryl groups. They are relatively unreactive and often used as solvents.

11. Aldehydes and ketones

Aldehydes and ketones contain the carbonyl group. Aldehydes have the carbonyl at the end of the chain, while ketones have it within the chain.

Reactions

- Nucleophilic addition.
- Oxidation of aldehydes.
- Reduction to alcohols.
- Condensation reactions.

12. Carboxylic acids and derivatives

Carboxylic acids contain the carboxyl group and are acidic because the conjugate base is resonance-stabilized. Their derivatives include esters, acid chlorides, amides, and anhydrides.

Important reactions

- Esterification.
- Neutralization.
- Decarboxylation.
- Nucleophilic acyl substitution.

13. Amines and amides

Amines are derivatives of ammonia and may be primary, secondary, or tertiary. They are basic due to the lone pair on nitrogen. Amides are less basic because resonance reduces availability of the lone pair.

14. Biomolecules

Biomolecules are organic molecules essential for life. They include carbohydrates, lipids, proteins, nucleic acids, vitamins, and enzymes.

15. Carbohydrates

Carbohydrates are polyhydroxy aldehydes or ketones, or compounds that yield such units on hydrolysis. They are the main energy source in living systems.

Classification

- Monosaccharides.
- Oligosaccharides.
- Polysaccharides.

Examples

- Glucose.
- Fructose.
- Sucrose.
- Starch.
- Glycogen.
- Cellulose.

Important reactions

- Oxidation.
- Reduction.
- Glycoside formation.
- Fermentation.
- Hydrolysis.

16. Lipids

Lipids are a diverse group of hydrophobic biomolecules including fats, oils, phospholipids, and steroids. They provide energy storage, membrane structure, and signaling roles.

Types

- Simple lipids.
- Compound lipids.
- Derived lipids.

Importance

- Energy reserve.
- Cell membrane formation.
- Hormone precursors.
- Thermal insulation.

17. Proteins

Proteins are polymers of amino acids linked by peptide bonds. They are the most versatile biomolecules and perform structural, enzymatic, transport, and regulatory functions.

Structural levels

- Primary structure.
- Secondary structure.
- Tertiary structure.
- Quaternary structure.

Denaturation

Protein structure can be disrupted by heat, pH change, or chemicals without breaking peptide bonds.

18. Nucleic acids

Nucleic acids store and transmit genetic information. DNA and RNA are polymers of nucleotides.

Nucleotide components

- Pentose sugar.
- Nitrogenous base.
- Phosphate group.

Functions

- DNA stores hereditary information.
- RNA participates in protein synthesis.

19. Enzymes

Enzymes are biological catalysts, mostly proteins, that increase the rate of biochemical reactions.

Properties

- High specificity.
- Lower activation energy.
- Sensitive to temperature and PH.
- Often require cofactors.

20. Vitamins

Vitamins are organic compounds required in small amounts for normal metabolism. They may act as coenzymes or precursors of coenzymes.

Classes

- Water-soluble vitamins: B-complex, C.
- Fat-soluble vitamins: A, D, E, K.

21. Chemical importance of biomolecules

Biomolecules are the basis of life because they store energy, build structures, catalyze reactions, and encode genetic information. Their chemistry explains metabolism, heredity, nutrition, and disease.

22. Summary

Organic chemistry is the study of carbon compounds and their reactions, with aliphatic hydrocarbons forming the hydrocarbon base, functional groups controlling reactivity, and biomolecules linking chemistry to life. This makes the chapter fundamental for both theoretical chemistry and biological applications.